



Module 20

Running Shoes, Apparel and Gear

Running shoe and apparel selection has a large influence on a runner.

The primary areas of influence are:

1. **Safety**
2. **Performance**
3. **Comfort**

❖ OBJECTIVES

Upon completion of this module, you should have an understanding of the following areas:

- Factors in selecting apparel
- Recommended apparel inventory
- Three primary areas that equipment influences
- Training assessment tools
- Shoe selection/characteristics
 - Including barefoot/overcushioned shoes
- Shoe fit/replacement guidelines
- Shoe cushioning: effect on performance

❖ RUNNING SHOES

There are two primary classifications of shoes: *conventional* and *minimalist*. However, there is also a recent trend of running barefoot and in overcushioned shoes.

Conventional shoes are designed to provide support and stability. Minimalist shoes are designed to allow the feet to use their inherent strength to support a runner instead of relying on the structure of a shoe. Minimalist shoes are typically used in conjunction with a midfoot strike.

HEEL-TO-TOE DROP



Figure 20.1 Heel-to-toe drop

This relates to the difference in the height of the sole at the heel and forefoot (in millimeters). Because of its padding and support, a conventional running shoe typically has a drop of 10 or more millimeters whereas a minimalist shoe typically has a drop of less than 10mm (364). Heel-to-toe drop has become a focal point with the increasing popularity of minimalist shoes. Individuals converting to or

currently running in minimalist shoes tend to look for shoes that have a small to zero heel-to-toe drop. While this level of heel-to-toe drop might be fine for some, it is not advised for those converting to a midfoot strike or for individuals with excessively tight calves and/or Achilles tendons. Successful progression to a minimalist shoe often takes time.

As noted previously, during a midfoot stride, the ball of the foot impacts the ground first, but almost immediately thereafter, the midfoot and heel lightly impact the ground to provide support. Assuming your client runs with correct form, a small- to zero-drop shoe will place extra eccentric stress on the calves and Achilles tendon. This is because without an elevated heel, the ankle is forced

to dorsiflex more. Therefore, you need to factor in ankle flexibility (flexion) with respect to your client's shoe selection.

In figure 20.1, the heel drop corresponds to A minus B, in millimeters.

Note: A *zero-drop* minimalist shoe has the same sole height at the heel and forefoot.

CONVENTIONAL SHOES

Conventional running shoes are geared toward runners who heel strike. More specifically, conventional running shoes primarily focus on the following three areas:

- Cushioning: primarily through midsole EVA foam
- Control rear foot motion
- Arch support



Figure 20.2 Conventional running shoe

The type of foot arch often (but not always) correlates to the amount of rear foot motion. Below is a chart of these correlations:

Arch Type	Typical Rear Foot Motion	Shoe Type Often Recommended
Low	Substantial Pronation	Motion Control
Normal	Mild Pronation	Stability/Neutral
High	None or Slight Supination	Neutral

Additionally, conventional running shoes typically have some or all of the following characteristics:

- Substantial heel padding
- Large heel-to-toe drop
- Relatively rigid shoe
- Heavy

The results of most studies regarding the impact of shoe type (e.g., stability, motion control, neutral) in respect to controlling foot motion are conflicting. Some studies show that the type of running shoe does not influence rear foot motion (197, 198), or it does so minimally, whereas other studies show that rear foot motion can be substantially affected by the type of shoe (199).

It is well-known that the body is able to adapt to change. Studies have demonstrated that runners are able to adapt to changes in the type of shoe as well as the type of running surface (200). This tends to support the argument that the specific shoe type minimally affects rear foot motion because of the body's ability to adapt to varying shoe types. Adaptability is one of the main reasons behind the barefoot/minimalist shoe movement. Those who run barefoot or in minimalist shoes often do so with the belief that conventional shoes take over the stabilization and support aspects that the body should naturally be providing. This, in turn, makes for a weaker foot and ankle complex.

Many shoe/foot studies are done with the feet in a static position or with runners who run for short bouts and therefore for whom fatigue is not a factor. It has been proposed that most injuries occur when a runner is fatigued and has reduced levels of muscle strength to control the ankle joint (199). Therefore, it could be deduced that a decrease in the body's ability to stabilize the ankles and feet would place greater reliance upon a shoe's structure for support.

While it is standard to recommend a specific type of running shoe based on foot type, according to a 2009 study by Richards et al., only minimal evidence exists to support that this practice reduces injury rates (201). Additionally, Richards et al. found no randomized, controlled trials that correlated the prescription of shoe type to injury rates. Richards et al. concluded that prescription of shoe type based on foot type is not an evidence-based practice (201).

Perhaps the most interesting study to support the findings of Richards et al. is a 2009 study by Knapik et al. This study examined the correlation of shoe

prescription and injury risk among new military trainees (195). Knapik et al. found that regardless of a trainee's static foot type (high/low arch), individual shoe prescription based on foot type did not lower injury risk compared with when all the trainees ran in the same type of shoe (stability).

The one caveat to these studies is that the participants ran low mileage relative to distance runners who typically run long miles. For example, participants in Knapik's study ran between .5 to three miles at a time. Therefore, more research is needed on high-mileage runners to adequately assess the relationship between shoe type and injury risk (202).

MINIMALIST SHOES



Figure 20.3 Minimalist running shoe

Characteristics

- Minimal heel padding
- Small heel-to-toe drop (typically 10mm or less)
- Minimal arch support
- Flexible sole and shoe fabric ("uppers")
- Light
- Wide toe box to take into account the expansion of the forefoot during impact

Unless your clients specifically ask for a minimal shoe at a running store, more often than not the staff will watch them run and put them in a

conventional shoe based on their visual assessment of rear foot motion (e.g., neutral, motion control, stability).

Most track- and cross-country-specific racing shoes are minimalist in design. Some of these shoes have small spikes on the sole for traction on the track and off-road, respectively.

Hybrid shoes fall in between minimal and conventional shoes. They are characterized by a medium level of heel-toe drop, heel padding, and weight and arch support.

BAREFOOT

If your client wants to run barefoot, the most important thing to realize is that not only does the body need to adapt biomechanically, but skin also needs to become more calloused. Walking around indoors in bare feet is the starting point to toughen up the feet, then walking outside in an area that is void of any debris that could puncture the skin. Prior to running outside, running on a treadmill in bare feet (if allowed) or on a running track is also helpful in the adaptation process.

Following are several reasons why people believe that running barefoot is more advantageous than running in shoes:

- Direct power transfer
- It is the way the body was biomechanically meant to run
- Less mass (i.e., shoe) to accelerate/decelerate equates to increased efficiency
- Strengthens the foot arch

When beginning to run barefoot outside, it is strongly advised to use shoes such as *Vibram Five Fingers*®. These are essentially a foot glove with a rubber tread on the bottom. The standpoint of this certification is that if your client wants to run barefoot outside, it should be done in shoes such as *Vibram*'s. This is because the chance for injury, and thus infection, due to glass or other debris penetrating the skin on the sole of the foot is high when running barefoot. However, a client

who insists on running barefoot must know the risks and follow a proper adaptation process.

A 2009 study by Kerrigan et al. found that while conventional running shoes provide good support to the foot, they place greater stress on the ankles, knees, and hips compared with running barefoot (323). The study consisted of 68 young, uninjured, active runners who averaged 15 miles/week. The study theorizes that the reason for increased stress on the three aforementioned joints is because conventional shoes lessen the amount of pronation of the foot/ankle. This reduces the body's natural shock absorption properties via foot/ankle pronation, and the shock is then transferred to the three joints (hip, knee, ankle).

OVERCUSHIONED SHOES

In contrast to running barefoot or in minimalist footwear, overcushioned shoes (aka: maximal cushioned) have up to twice the cushioning of standard running shoes.



Figure 20.4 Overcushioned running shoe

The genesis for the creation of overcushioned shoes was to serve the ultra-marathon market (673). However, since this time, the key demographic has become broader to encompass runners of all ability levels and to apply to events of varying distances.

Because of their relative newness, currently there is no peer-reviewed research regarding overcushioned shoes. However, following are several things to consider (674):

- Many large running shoe companies make at least one model of overcushioned shoes.
- Overcushioned shoes can take away from one's proprioceptive interaction with the ground while running.
- Overcushioned shoes may decrease a runner's power transfer.
- Individuals who favor overcushioned shoes note the following benefits:
 - Extra cushioning feels good
 - Shoes seem to help with forward propulsion
 - Reduction of leg fatigue over longer distances and thus a shorter recovery period
 - Fewer injuries, specifically in regard to the lower leg

Individuals must run in overcushioned shoes before they decide if they are the right option for them. However, as overcushioned shoes decrease one's proprioceptive interaction with the ground, it is advised that new runners not run in them until they have increased proprioception as it pertains to running and effective mechanics and functional strength of the foot/ankle complex.

Despite their large appearance, many overcushioned shoes are relatively light.

Like conventional running shoes, the composition of the sole of an overcushioned shoe has a lot to do with how a shoe "feels" when running. A soft midsole may feel forgiving and soft but may also diminish one's proprioception of the ground and reduce the potential energy of a runner because of the sole-absorbing energy. Conversely, a sole that is denser may act to absorb shock and, at the same time, assist with forward propulsion because of the rebounding characteristics of the midsole. The rebounding aspect of a shoe will likely diminish the energy potential of the stretch-shortening cycle (SSC). This is because the push-off assistance one gets from a shoe reduces the degree of effectiveness of the SSC. As previously noted, there is not a lot of research in this area so it is currently unknown if the sum total of the assistance one gets from a shoe and the SSC in overcushioned shoes is greater or less than when running in traditional or minimalist shoes.

Independent of the energy discussion, runners who are injury prone (especially lower legs) may realize a reduction in their injury rate through the use of overcushioned shoes. Like all aspects of training, introduction to a new running shoe type requires a progressive adaptation.

EFFICIENCY

A 2012 study by Kram et al. found that running in cushioned shoes was more efficient in regard to VO_2 and metabolic power than running barefoot (326). This is because there is an increased metabolic cost with running barefoot. While running in shoes does increase the metabolic cost by adding mass to a runner, it is hypothesized that the cushioning cancels out the metabolic cost (327).

A study by Logan et al. researched the difference in ground reaction forces (GRF) between three different types of running shoes: cushioned, racing flats, and racing spikes (609). **Ground reaction force** pertains to the force that the body exerts on the ground (e.g., impact). The primary finding was racing flats and racing spikes have a substantially higher GRF and loading rate than cushioned shoes.

The **loading rate** relates to the speed at which GRF occurs – in other words, the speed at which loads are applied to the body. While minimally cushioned shoes such as racing flats and spikes likely enhance a runner's efficiency from a biomechanical standpoint because of a direct power transfer (e.g., minimal energy lost through the absorption of shoe cushioning), it is likely that the metabolic cost is much higher than with a more cushioned shoe and possibly puts a runner at an increased risk of injury (609). Because of the increased GRF and loading rates of racing flats and spikes, it could be theorized that this type of footwear would be most applicable to shorter-distance events.

ARCH SUPPORT DEBATE

Following are common theories as to why individuals are for and against arch support.

For Arch Support

- Favors orthotics and shoes that offer arch support. Shoe types that provide arch support are *stability* and *motion control* shoes.

- An individual's arch is not strong enough to support the foot when running. Therefore, orthotics and/or arch-supporting shoes are needed to compensate for the lack of inherent arch strength/support.
- Overpronation can lead to injury and therefore needs to be lessened or eliminated via support shoes or orthotics.

Against Arch Support

- The running style most associated with a need for arch support is a heel strike, which is biomechanically incorrect. The correct form is a midfoot strike.
- Pronation is normal and, therefore, does not need to be corrected (stabilized).
- Arch support weakens the arch. Conversely, shoes without arch support strengthen the arch.
- As the longitudinal foot arch produces elastic energy when running, arch support greatly reduces this energy source, thereby reducing the efficiency of a runner.
- Arch support reduces pronation, which acts as the body's natural shock absorber. This, in turn, increases stress in lower-extremity joints (323).

RUNNING SHOE FIT/REPLACEMENT GUIDELINES

Regarding shoe fit, the most important thing is to try on shoes with the same socks and orthotics (if used) as used for training and racing.

Correct Fit

Front of Shoe: There should be at least half a thumb's width but not more than a full thumb's width between the longest toe and the front of the shoe. Running for long periods as well as in hot weather causes the foot to expand. Selecting a shoe that is too small will likely cause the foot to become cramped when running for long periods and/or in hot weather.

A shoe should fit snugly, but not tight in any one spot. The heel should not feel like it is going to come out of the shoe at any point.

Breaking in Shoes

While there is almost no break-in period with running shoes today because of their relaxed structure (203), this does not mean that the specific design and nuances of a shoe might not take getting used to. Shoes should feel comfortable when trying them on. If your clients experience pain or the shoe just doesn't feel right in the store, they should not purchase them with the hope that they will get more comfortable once broken in. It is advised that your clients wear new shoes around the house or while running errands prior to running in them. When running in new shoes initially, it is ideal to start with shorter distances to see how one's feet respond to the shoes. For example, perhaps the shoes fit great but the tongue moves to the side while running and puts pressure on the side of the feet. This is not something your client wants to discover on a long training run.

Seeing how shoes feel when running in them is important. Many running stores have treadmills for this purpose, and some stores allow customers to run around the block. Running outside is ideal, but if this is not an option, a treadmill or running in the store is fine. Some running stores also allow customers a long return window (i.e., 10 days) so they can take them home and run in them to know if they fit properly.

Replacement Guidelines

Depending on the source, running shoes should be replaced between 400 and 600 miles (203). While this range is sufficient, three key variables ultimately should be the determining factor in shoe replacement:

1. Comfort
2. Fit
3. Shoe Condition

If any of these factors are substantially compromised, the shoes should be replaced – regardless of age or mileage. Continued use of shoes that are deficient in any of the aforementioned areas would at best diminish performance and, at worst, cause injury.

Shoe Breakdown's Affect on Biomechanics

While it is well-known that running shoes lose some of their cushioning over time, what is less known is how the loss of cushioning affects a runner's form.

In a 2009 study by Kong et al., 24 runners (14 men, 10 women) were divided into three groups, each group wearing a different type of cushioned shoe. Over the course of the study, the subjects ran 200 miles (200). Runners were assessed biomechanically before and after the 200 miles. The three types of shoes used in regard to cushioning were air, gel, and spring-cushioned.

After 200 miles of running, there were only very small changes in subjects' running mechanics and, interestingly, there were no reported changes in impact forces. So, what does this mean? This information seems to support the notion that the body adapts to changes in foot cushioning and support. As decreases in shoe cushioning and support occur, the body adapts to maintain proper running mechanics. Additionally, there was no difference in running mechanics from one type of shoe cushioning to the next. This finding relates to the effectiveness of different types of shoe cushioning. Findings from the Kong et al. study demonstrate that regardless of shoe brand and shoe cushioning type (i.e., air, gel, spring), the primary source of cushioning is midsole EVA foam, and with respect to EVA midsole foam, there is no discernible difference between shoe brands (200).

Shoe Compression

The exact percentage of cushioning lost in relation to miles run is unknown as there are many variables (e.g., running style, runner's weight, etc.). However, it has been theorized that the largest amount of cushion loss occurs within the first 200 miles (204). It is important to stress to clients that while they do not want to

run in a shoe that is uncomfortable, over time they should not expect shoes to feel the same as when they were brand new.

A common practice among distance runners is to swap out identical pairs of running shoes during training. Generally speaking, this is a good idea, as it will provide more cushioning for a client during runs. Another common reason for doing this is to “rest” or allow shoes to “decompress” before the next run. The standard advice given in this area is to let shoes rest between 24 and 48 hours before running in them again, the theory being that it takes this long for the EVA midsole foam to return to its decompressed state. A study by Cook et al. found no difference in the retention of cushioning between shoes that were rested and those that were not, when used within a 24- to 48-hour period (204).

Shoe Cushioning: Effect on Performance

According to a study by Kram et al., the degree of foot/shoe cushioning in respect to efficiency resembles a bell curve (327). Too much or too little cushioning decreases efficiency (metabolic power). This study had 10 barefoot runners run with a midfoot strike on a rigid treadmill (no shock absorption) using the following scenarios:

1. Barefoot
2. Barefoot on treadmill that was covered in 10mm EVA foam
3. Barefoot on treadmill that was covered in 20mm EVA foam

Note: 10mm is roughly the same height of cushioning in the forefoot of lightweight running shoes

The conclusion: Running with 10mm of EVA foam required approximately 2 percent less energy than running barefoot, while running with 20mm of EVA foam required 1.7 percent less energy than running barefoot.

❖ ASSESSMENT TOOLS

The sport of running can be quite technical in regard to biomechanics and cardiovascular performance. Fortunately, several training assessment tools can provide quantitative measurements and establish benchmarks.

GENERAL

Many of the general assessment tools listed below have been noted previously in the certification.

Body Fat Calipers: Used to measure an individual's body fat percentage. Calipers are typically spring-loaded and measure body fat in millimeters.

Bioelectric Impedance: Used to measure body fat percentage via a small, harmless electrical current. Typically are either handheld devices or scales.

Goniometer: Essentially a large protractor with moving arms, it is used to measure angles of the body.

Blood Lactate Analyzer: Typically a handheld device that requires a small blood sample to calculate blood lactate level.

Gulick Tape: Used to take girth measurements, this is a measuring tape with a small spring attached to the end to ensure consistent tape tension.

Laser Level: Used to perform postural assessments.

Stop Watch: Used for performing timed assessments.

RUNNING

A heart rate monitor (HRM) is a popular assessment tool used by runners. How data from heart rate training is used ultimately determines its effectiveness.

Although basic, the stopwatch feature on watches is used a lot by runners to measure total run time and record splits (if running repeats).

A *GPS* (global positioning system) device enables runners to accurately determine their running pace in real time.

Motion-capture software and video cameras are helpful in analyzing a runner's gait to assess stride and foot/ankle movement. It is useful to show clients their running form so they understand how and why corrections are being made.

Last but not least, the most inexpensive training assessment tool is the training log. This tool is beneficial in many areas:

- Keep athletes motivated and on track.
- If an injury occurs, a log might help explain the reason (e.g., sudden increase in distance).
- Track data: heart rate, distance, how client felt, missed workouts, intensity/time.
 - Historical data represents the true value of a training log, as a coach is able to see trends and progress.

A training log is a “must-have” for your clients. An accurate log will help you be a more efficient and effective coach.

❖ FACTORS IN SELECTING APPAREL

Proper selection of clothing is not a function of only comfort but also safety and performance. Fabric materials and clothing design are always advancing to reflect the increasing demands of endurance athletes.

There are several factors to consider when selecting clothing:

- Fabric composition (wicking fabrics)
- Proper fit
- Clothing for all types of weather
- Function
- Comfort

WICKING FABRICS

Wicking refers to the ability of a fabric to “breathe.” Unlike fabrics such as cotton that absorb sweat, wicking fabrics allow body heat and sweat to pass through so the body stays cool and dry.

PROPER FIT

Proper fit of clothes is important for performance clothing. Depending on the preference of a runner, clothing might fit snugly or be more relaxed.

CLOTHING BASED ON THE WEATHER

An athlete must be ready for all types of weather. This means it is imperative to have a wide selection of clothing. Not having a proper selection of clothing compromises not only an athlete's comfort, but also performance and safety.

Cold Weather

A base layer (layer next to skin) that provides warmth and wicks sweat away from the body is a necessity when exercising during cold weather. The colder the weather, the more layers are needed. For cold and windy days, it is advised to have some sort of windproof outer layer so that wind chill is minimized. Following is a chart of cold-weather running clothing in respect to applicable temperatures. A few things to keep in mind:

- Individuals respond differently to hot and cold weather; therefore the below chart is only rough guidelines.
- A clothing product will most likely have varying insulation properties depending on the purpose. For example, tights come in varying thicknesses based on the temperature for which they are designed.
- Many cold-weather clothing items indicate on the tag what range of temperatures they are recommended for.
- It is better to overdress and shed layers than underdress.
- Dark-colored clothing will absorb more heat than light-colored clothing.
- Normally, less clothing is worn during a race than in training, as the effort level is usually greater and thus more body heat is being generated.
- It is important to wear sunscreen on any exposed skin, regardless of the time of year.

RUNNING

<u>Item</u>	<u>Temperature</u>
Arm Warmers	<45 (7.2 C)
Thin Tights	<45 (7.2 C)
Thermal Tights	<35 (1.7 C)
Vest	<45 (7.2 C)
Long-Sleeve Base Layer	<35 (1.7 C)
Long-Sleeve Shirt	<55 (12.8 C)
Wind Jacket	<50 (10 C)
Insulated/Thermal Jacket	<30 (-1.1 C)
Thin Finger Gloves	<45 (7.2 C)
Insulated/Thermal Gloves	<35 (1.7 C)
Full Face Mask	<25 (-3.9 C)
Hat	<40 (4.4 C)
Winter Running Socks	<40 (4.4 C)

Note:

The temperatures factor in wind chill

Guidelines

While a wide selection of clothing options make training and racing in most conditions possible, it can be challenging to figure out which pieces of clothing to wear. A few rough guidelines can help with making these decisions:

1. Always dress in layers.
2. In cold weather, an athlete should be somewhat cold for the first five to 10 minutes when running.
3. In cold weather, the more intense the training or racing is, the fewer layers that should be worn.
4. Always try out different clothing options during training before racing in them.

❖ APPAREL INVENTORY

Following are common types of running apparel for training and racing:

- Running top (sleeveless/short sleeve/long sleeve)
- Jacket
- Sports bra
- Hat/Visor

- Headband
- Vest
- Shorts
- Tights
- Arm warmers
- Gloves
- Socks/Compression socks
- Sunglasses

Some areas to consider when selecting apparel are:

- **Seams:** Make sure that the seams will not rub skin in an abrasive manner.
- **Pockets:** What and how much do you need to carry?
- **Reflective clothing:** Never a bad idea and mandatory for low light conditions.
- **Fit:** Make sure clothes don't chafe during training before using them in races.
- **Brand:** Don't assume that just because a piece of apparel from a brand fits well that all other pieces in the same size from the same brand will also fit well.
- **Keys:** Will you be carrying your keys when running? Key pockets that attach to shoe laces may be a good option.

COMPRESSION SOCKS/CALF SLEEVES

Many runners wear compression socks and compression calf sleeves. Compression socks cover the foot and extend to the top of the calf. Compression sleeves slide on over the calf and just cover the calf.

The science behind compression socks comes from clinical settings where they are used to treat edema (buildup of fluid beneath the skin) and other medical issues. The theory behind compression socks is to increase the rate of blood flow back to the heart (venous return) and to reduce blood pooling in the legs. Another reason why some people use them is to reduce the vibration of the calf muscles, thereby decreasing fatigue.

A 2011 study out of Stellenbosch University in South Africa sought to determine the effectiveness of compression socks in post-exercise recovery for male long-distance runners (268). The study assessed a total of 83 runners competing in a

56-km (35-mile) ultra-endurance run. Following is a summary of the findings regarding the effectiveness of compression socks:

- Runners wearing compression socks had less muscle damage post-race than those without compression socks
- Those wearing compression socks exhibited less calf swelling
- Some compression socks may not provide enough compression to elicit the desired results
- Theorized that compression socks provide dynamic stability to the calf (i.e., flexible cast)
 - Calf stability reduces muscle vibration, which is theorized to increase muscular endurance and reduce soreness and muscle damage
- Benefits are limited to calf muscles (gastrocnemius and soleus) via reduced structural damage
- Increased cellular repair
- Increased venous return is inconclusive

These findings are substantiated by several other studies (265, 266, 267).

While the preceding data seem to show that compression socks may decrease muscle damage (calf) while running and improve recovery, another study found that compression socks did not increase running performance. A 2015 study by Stickford et al. found that lower-leg compression sleeves did not improve an individual's running mechanics (ground contact time, step frequency, and length) or VO_2 max. In other words, the running economy of test subjects was not altered using calf compression sleeves (675).

Lastly, a 2014 study by Hill et al. examined the effect of calf-compression apparel in respect to accelerating recovery post-marathon (689). There was no physiological recovery improvement in the group that wore compression socks over the group that did not. Interestingly, the group that wore compression socks noted a significantly lower level of perceived muscle soreness (after 24 hours) as compared with the group that did not wear compression socks. As the use of compression socks improved the sense of recovery without actually improving recovery from a physiological standpoint, the benefit of compression wear may be more psychological than physiological.

Conclusion

The physiological benefit of wearing compression socks/sleeves during and after a run is largely inconclusive, with the most recent data leaning toward the apparel showing no benefit. However, because of potential psychological benefit and the fact there is no physiological downside to wearing compression socks, if a client wants to wear them while running and for recovery, it should not be discouraged.

❖ OTHER RUNNING GEAR

- Smartphone/*iPod* arm holder
- Hydration belt
- Handheld water bottle
- Anti-chafe balm/spray
- Backpack
- Jogging stroller
- Foam roller
- Headlamp (if running at night)
- Sweat-proof sunscreen
- GPS watch
- Waist pack/*SPIbelt*™

❖ SUMMARY

- The major factors in apparel selection are:
 - Fabric composition (wicking fabrics)
 - Proper fit
 - Clothing for all types of weather
 - Function
 - Comfort
- In cold weather, clothing should be layered.
- Compression socks can assist in recovery and may improve running performance.
- It is important to have a wide range of clothing options so that a client is prepared for any weather condition.
- While EVA foam in running shoes compresses over time, allowing shoes 24–48 hours to “decompress” before running in them again is a myth.
- Conventional running shoes are characterized by substantial heel padding, cushioning, and arch support.
- Minimalist shoes are characterized by minimal heel padding, cushioning, and arch support. Minimalist shoes have a low heel-to-toe drop.
- Overcushioned shoes may be a solution for those who are injury prone, specifically in the lower legs.
- Shoe cushioning may assist in forward propulsion but likely diminishes the effect of the stretch-shortening cycle (SSC). A large heel-to-toe drop also reduces the SSC of the Achilles tendon.
- There is little to no break-in time required with today’s running shoes.